

R PROGRAMMING

Study Hours and Exam Scores

Exploratory Data Analysis Project

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Subject	R programming
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10-Student Cohort • Exploratory Analysis Report

1. INTRODUCTION & DATA INGESTION

This report examines the relationship between the number of hours students spend studying and their final exam scores. The objective is to explore whether study time can act as a useful indicator of academic performance within a small 10-student sample cohort.

R DATA INGESTION

R code	<code>student_data <- read.csv("student_data.csv")</code>
Structure	<code>str(student_data)</code>

2. COMPLETE DATASET RECORD

#	Name	Gender	Study Hours	Exam Score	Result
1	Aarav	Male	2.5	42	Fail
2	Diya	Female	4.0	55	Pass
3	Kabir	Male	5.0	61	Pass
4	Meera	Female	6.5	72	Pass
5	Arjun	Male	3.5	50	Fail
6	Ishita	Female	7.0	78	Pass
7	Vivaan	Male	8.0	84	Pass
8	Anaya	Female	5.5	68	Pass
9	Reyansh	Male	9.0	91	Pass
10	Kiara	Female	9.5	95	Pass

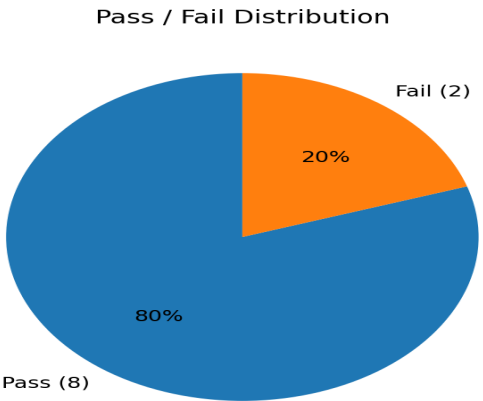
3. DATA PREPROCESSING & STATISTICAL SUMMARY

Data quality was checked before analysis. The dataset contains 10 complete student records, with no missing values in the variables used for the study. Study hours were treated as numeric values and exam scores were treated as quantitative performance measures.

Mean Exam Score 69.6	Mean Study Hours 6.0 hrs	Std. Dev. (Score) 17.71	Pass Rate 80%
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R summary calculations: `sd(student_data$ExamScore) = 17.71`; Female = 5 and Male = 5; Pass = 8 and Fail = 2.

4. RESULT DISTRIBUTION

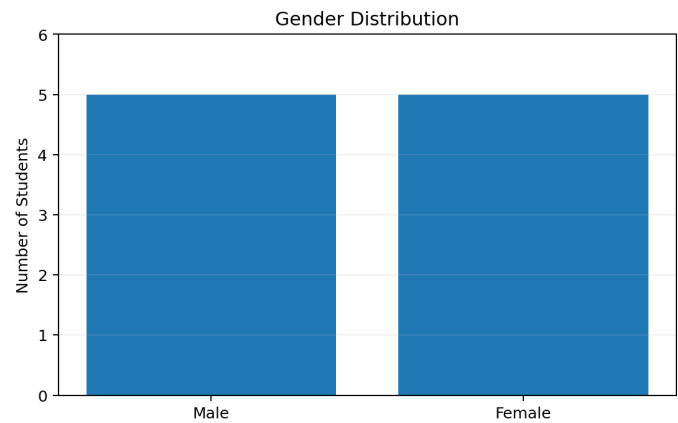


The sample contains 8 passing outcomes and 2 failing outcomes, corresponding to an 80% pass rate.

4. DEMOGRAPHIC & OUTCOME DISTRIBUTIONS

Categorical Frequency Distributions

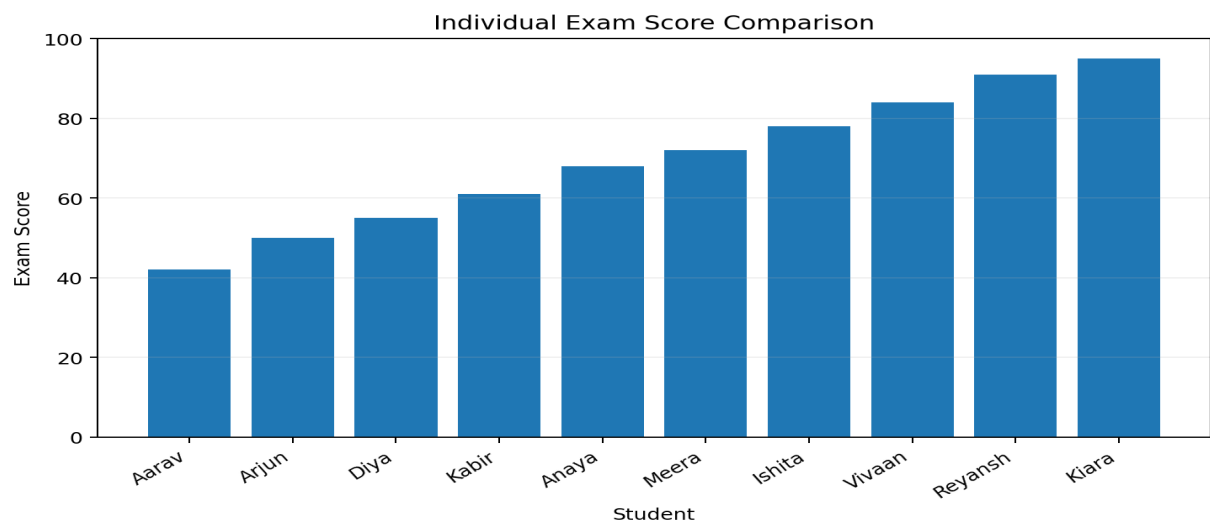
Category	Count
Male	5
Female	5
Pass	8
Fail	2



The gender distribution is balanced in this dataset, with five male and five female students.

5. INDIVIDUAL PERFORMANCE COMPARISON

The records show a general upward pattern: students with more study hours tend to have higher scores.



5. INDIVIDUAL PERFORMANCE COMPARISON

Student	Study Hours	Exam Score	Observation
Aarav	2.5	42	Lower score range
Diya	4.0	55	Moderate score range
Kabir	5.0	61	Moderate score range
Meera	6.5	72	Moderate score range
Arjun	3.5	50	Lower score range
Ishita	7.0	78	Higher score range
Vivaan	8.0	84	Higher score range
Anaya	5.5	68	Moderate score range
Reyansh	9.0	91	Higher score range
Kiara	9.5	95	Higher score range

Visual interpretation

The lowest study-hour records (2.5–3.5 hours) correspond to the two failing outcomes, while the highest study-hour records (9.0–9.5 hours) correspond to the two highest exam scores in this sample.

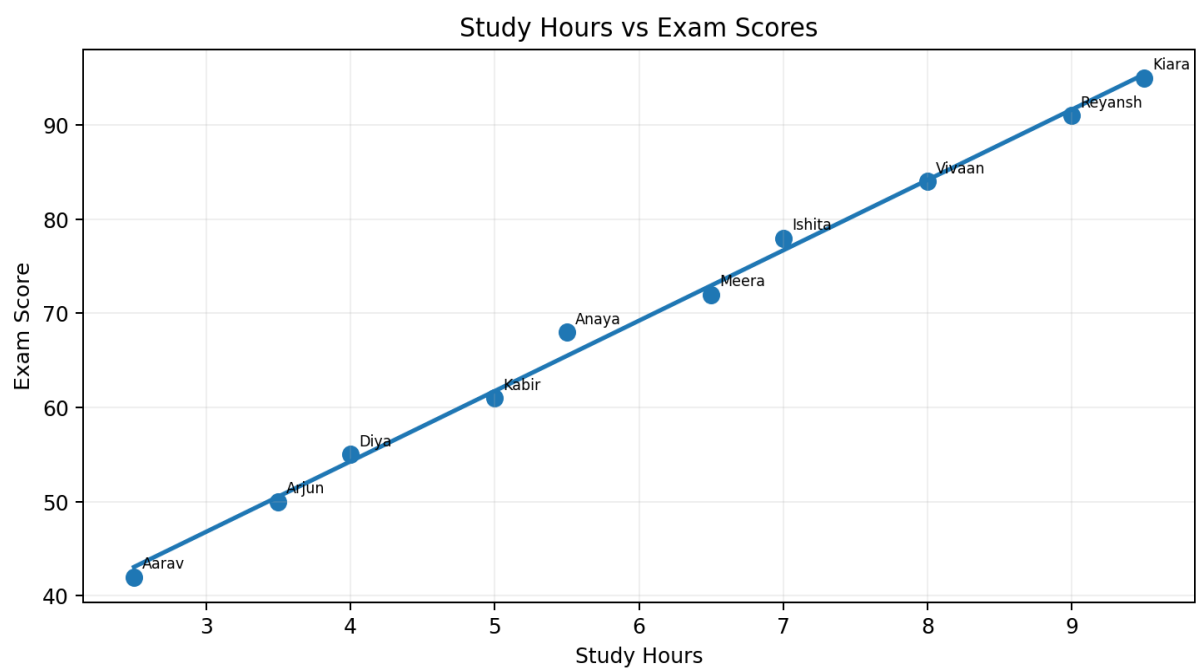
The individual comparison chart on the previous page provides a direct visual view of score differences across all ten students.

6. BIVARIATE CORRELATION & REGRESSION ANALYSIS

To quantify the linear association between study hours and exam scores, Pearson's correlation coefficient was calculated and a simple linear regression model was fitted.

R CORRELATION & REGRESSION

R code	<code>correlation_coeff <- cor(student_data\$StudyHours, student_data\$ExamScore)</code>
R code	<code>fit <- lm(ExamScore ~ StudyHours, data = student_data)</code>
Result	<code>correlation_coeff = 0.9979</code>
Model	<code>ExamScore = 24.34 + 7.48 × StudyHours</code>



Scatter-plot interpretation: Pearson's $r = 0.9979$ indicates a very strong positive linear association in this sample. The regression slope is approximately 7.48 score points per additional study hour.

7. FINDINGS & ACTIONABLE RECOMMENDATIONS

Key Analytical Takeaways

- Strong Positive Association:** The calculated Pearson coefficient ($r = 0.9979$) shows that study hours and exam scores move closely together in this 10-student dataset.
- Lower-Study Group:** The two students who did not pass recorded 2.5 and 3.5 study hours, suggesting that very low study time may be associated with weaker outcomes in this sample.
- High Achievers:** The two highest scores, 91 and 95, were recorded by students studying 9.0 and 9.5 hours respectively.
- Practical Recommendation:** Students can use scheduled, consistent study sessions and monitor their study time alongside practice-test scores. However, this small observational dataset shows association rather than proving that study hours alone cause higher exam scores.

CONCLUSION

Within this 10-student cohort, higher study hours are strongly associated with higher exam scores. The analysis supports using study-time tracking as one useful academic monitoring indicator, while recognizing that factors such as study quality, prior knowledge, attendance, and learning methods may also affect exam performance.

Project summary	10 students
Mean study time	6.0 hours
Mean exam score	69.6
Correlation	0.9979
Pass rate	80%

End of Project Report